Test Report — Take Profit / Stop Loss

Leverage Trading with Market/Limit Orders on Minswap · Project 1400078 · Milestone 3

Subject	Take-profit and stop-loss in the leverage protocol's Plutus validators, and their automatic execution on Cardano preprod
Source	github.com/dthuongbk/catalyst · leverage trading/smart-contract at commit 5ee9812
Toolchain	Aiken v1.1.17, Plutus V3
Network	Cardano preprod, the deployment recorded in DEPLOYMENT.md
Date	2026-11-04

Result. 22 of 22 TP/SL unit test cases pass, and the full suite passes 291 of 291. All 4 on-chain gate cases behave as expected, and 5 end-to-end TP/SL flows on preprod closed and repaid their loans automatically.

1. Scope

- Take-profit and stop-loss prices are stored in the order and position datums and validated when they are written.
- The trader can set, change or clear them on an open position; nobody else can.
- The close is accepted without the trader's signature only when the oracle rate, built from Minswap pool state, has reached a threshold: at or above the take-profit price, at or below the stop-loss price.
- The close repays the loan to the Danogo Flexible Lending Pool and leaves the remainder for the trader to collect.

2. Method

Unit tests run with `aiken check`. Each test calls a validator directly with a constructed transaction. Every rejection case is an accepting fixture with one field changed, so a rejection is attributable to that field. On-chain cases were run on preprod against the deployed validators and a live oracle rate, and end-to-end flows were read back from the chain.

3. Unit test cases

ID	Test	Expected	Result	Memory	CPU steps
Thresholds stored in the datum					
U01	`tests/position_utils_test valid_position_params_bounds_each_field`	TP and SL accepted when unset (0) or positive; rejected with a negative numerator or a zero denominator	**PASS**	343,798	117,250,218
U02	`tests/create_position_test create_position_rejects_a_datum_that_fails_the_param_bounds`	A position cannot open with a zero take-profit denominator	**PASS**	443,065	145,683,376
U03	`tests/execute_order_test execute_rejects_a_position_datum_that_is_not_a_copy_of_the_order`	A fill must copy the order's terms, TP and SL included, into the position	**PASS**	1,183,338	378,136,836
U04	`tests/wire_format_test datum_arities_are_the_deployed_ones`	OrderDatum has 14 fields and PositionDatum 6, as the deployed contract reads them	**PASS**	195,609	60,274,265
U05	`tests/wire_format_test datum_fields_sit_where_the_deployed_contract_reads_them`	Fields after stop_loss_price sit at the index the deployed contract reads	**PASS**	106,351	31,388,047

Setting and editing TP/SL on an open position					
U06	`tests/modify_position_test` `modify_position_accepts_a_new_take_profit_price`	A new take-profit price is accepted	**PASS**	713,515	223,988,880
U07	`tests/modify_position_test` `modify_position_accepts_clearing_the_take_profit_price`	Setting take-profit to 0 turns it off	**PASS**	713,515	223,988,880
U08	`tests/modify_position_test` `modify_position_accepts_a_new_stop_loss_price`	A new stop-loss price is accepted	**PASS**	714,711	224,402,055
U09	`tests/modify_position_test` `modify_position_accepts_clearing_the_stop_loss_price`	Setting stop-loss to 0 turns it off	**PASS**	714,711	224,402,055
U10	`tests/modify_position_test` `modify_position_rejects_a_zero_take_profit_denominator`	A take-profit price with a zero denominator is rejected	**PASS**	569,281	178,430,443
U11	`tests/modify_position_test` `modify_position_rejects_a_zero_stop_loss_denominator`	A stop-loss price with a zero denominator is rejected	**PASS**	589,989	183,557,566
U12	`tests/modify_position_test` `modify_position_rejects_an_owner_input_without_the_ticket`	Only the holder of the PositionOwnerNFT can edit TP/SL	**PASS**	336,533	97,070,445
U13	`tests/modify_position_test` `modify_position_rejects_the_position_as_its_own_owner_input`	The position UTxO cannot stand in for the owner's ticket	**PASS**	357,804	104,009,158
U14	`tests/modify_position_test` `modify_position_rejects_a_new_long_token`	An edit cannot change the collateral token	**PASS**	547,609	169,362,703
U15	`tests/modify_position_test` `modify_position_rejects_a_new_max_slippage`	An edit cannot change the maximum slippage	**PASS**	568,507	174,834,037
U16	`tests/modify_position_test` `modify_position_rejects_a_new_beneficiary`	An edit cannot redirect the proceeds	**PASS**	569,053	177,689,195
Trigger rule					
U17	`loan/repay_loan` `standing_instructions_face_opposite_directions`	Take-profit fires at or above its price, stop-loss at or below its price, both inclusive (5 price cases)	**PASS**	221,998	94,341,418
Closing without the trader's signature					
U18	`tests/close_position_test` `close_position_burns_the_position_alone_where_no_owner_is_presented`	A TP/SL close burns the position and leaves the PositionOwnerNFT with the trader to collect the remainder	**PASS**	383,835	121,058,236
U19	`tests/close_position_test` `close_position_burns_both_where_the_owner_presents_the_ticket`	A close by the owner burns both NFTs	**PASS**	445,531	136,696,164
U20	`tests/close_position_test` `close_position_rejects_an_owner_input_with_a_minus_one_index`	An owner input together with the 'no owner' index is rejected	**PASS**	400,614	122,585,270

U21	`tests/close_position_test close_position_rejects_a_named_owner_index_with_no_owner_input`	An owner index with no owner input is rejected	**PASS**	358,701	111,848,538
U22	`tests/close_position_test close_position_rejects_a_missing_loan_withdrawal`	A close without the loan withdrawal, which carries the repayment to the pool, is rejected	**PASS**	359,466	114,778,578

Memory and CPU are the execution units each test consumed.

4. On-chain gate cases

The same closing transaction was evaluated on preprod with only the threshold changed (DEPLOYMENT.md, section 4.20).

ID	Case	Expected	Result
G01	Take-profit unset (0/1)	Close without the owner's ticket rejected	Rejected
G02	Take-profit at twice the oracle rate (not reached)	Rejected	Rejected
G03	Take-profit at half the oracle rate (reached)	Accepted	Accepted
G04	Stop-loss reached, take-profit out of reach	Accepted	Accepted — 0f4d7ef193509eeb…

5. End-to-end flows on preprod

Each flow was opened from a trader wallet. The trigger was submitted by the backend keeper, which never holds the trader's PositionOwnerNFT. The full transaction sequence of each flow is in `milestone3/evidences/Leverage_TPSL_transactionsHash.csv`.

ID	Flow	Expected	Result	Evidence
E01	Stop Loss SL 520/1, set at open	Keeper places the close without the owner's ticket; Minswap fills it; the loan is repaid	**PASS**	trigger 4119f18362f23624… repaid 68a30593800a62c2…
E02	Take Profit TP 370/1, set by an edit after open	Keeper places the close without the owner's ticket; Minswap fills it; the loan is repaid	**PASS**	trigger 20d4d8e4a9f4980a… repaid 3c8f91ecf6feaaec…
E03	Take Profit TP 230/1, set at open	Keeper places the close without the owner's ticket; Minswap fills it; the loan is repaid	**PASS**	trigger 73495544e8b57399… repaid 836257a3afbf77a5…
E04	Take Profit TP 228/1, set at open	Keeper places the close without the owner's ticket; Minswap fills it; the loan is repaid	**PASS**	trigger d178c8811830eaa7… repaid 22297ff331a405df…
E05	Take Profit TP 360/1, set by an edit after open	Keeper places the close without the owner's ticket; Minswap fills it; the loan is repaid	**PASS**	trigger 966c51a6bf56a70c… repaid 6ec41360691dc92c…

6. Resource usage on chain

Largest execution units and size per step across the TP/SL flows on preprod, as a share of the per-transaction limit, and the median fee.

Step	Txs	Memory	CPU steps	Size (bytes)	Median fee
Open position with TP/SL	14	7,299,545 (42%)	2,486,907,776 (25%)	7,144 (44%)	2.19 ADA
Set or edit TP/SL	2	724,575 (4%)	245,272,095 (2%)	2,204 (13%)	0.42 ADA
TP/SL trigger (keeper)	14	4,341,839 (25%)	1,480,008,507 (15%)	3,234 (20%)	1.62 ADA
Repay the loan to the pool	14	4,790,766 (27%)	1,666,607,347 (17%)	2,714 (17%)	1.61 ADA
Collect the remainder	2	1,332,618 (8%)	433,313,666 (4%)	3,095 (19%)	0.95 ADA

Limits: 17,500,000 memory units, 10,000,000,000 CPU steps and 16,384 bytes per transaction. The largest validator, `loan`, is 14,532 bytes.

7. Known limits

- After a trigger, the close waits for the Minswap batcher, which runs more slowly on preprod than on mainnet. The protocol does not control it.
- Gate cases G01–G03 are evaluation results, not submitted transactions: a rejected transaction never reaches the chain.